

ILLINOIS TECH | College of Computing

ITMM 485 / 585
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Legal and Ethical Issues in
Information Technology



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Ethical Concepts and Theories

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Ch12: Ethical Aspects of
Emerging and Converging
Technologies



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Emerging and Converging
Technologies

- Chapter 12 examines ethical aspects of three key emerging/converging technologies:
 - ambient intelligence (AmI)*,
 - nanocomputing*,
 - autonomous machines (AMs)*.
- This chapter also examines issues in the emerging field of *machine ethics*, and it describes a “dynamic” ethical framework for addressing challenges likely to arise from emerging technologies.

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Learning Objectives

Upon completion of this lesson the students should be able to:

- Explain “convergence” in the context of cybertechnology
- Define **ambient intelligence** and discuss social and ethical concerns that it raises
- Define **nanotechnology** and describe the ethical concerns associated with it
- Describe and discuss ethical concerns affecting **bioinformatics** research and development
- Define “**autonomous machines**” and discuss the ethical challenges they pose
- Discuss **Robotic ethics**
 - Recall Asimov’s Three Laws of Robotics
 - Describe and discuss the issues in creation of machines capable of making moral decisions

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Ethical Concepts and Theories

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Ch12: Ethical Aspects of Emerging
and Converging Technologies
PI: Convergence and AmI



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Module I Objectives

Upon completion of this lesson the students should be able to:

- Explain “convergence” in the context of cybertechnology
- Define **ambient intelligence** and discuss social and ethical concerns that it raises

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Converging Technologies and Technological Convergence

Howard Rheingold (1992) notes that technological convergence occurs when unrelated technologies or technological paths intersect or “converge unexpectedly” to create an entirely new field.

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Technological Convergence

- Convergence in the context of cybertechnology is by no+ means new or even recent
- Network technologies resulted from the convergence of computing and communications technologies in the late 1960s and early 1970s.
- Howard Rheingold notes that virtual-reality (VR) technology (examined in Chapter 11) resulted from the convergence of video technology and computer hardware in the 1980s.



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Converging Technologies and Pervasive Computing

- Currently, cybertechnology is converging with non-cybertechnologies at an unprecedented pace.
- For example, cyber-specific technologies are converging with non-cybertechnologies, such as biotechnology and nanotechnology.
- Cybertechnology is also becoming pervasive and ubiquitous as computing devices now permeate both our public and private spaces (in connection with ambient-intelligence-related technologies).

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1. Ambient Intelligence (AmI)

- Ambient Intelligence (AmI) is typically defined as a technology that enables people to live and work in environments that respond to them in “intelligent ways” (Aarts and Marzano, 2003; Brey, 2005; and Weber et al., 2005).
- Intelligent home
- IoT

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AmI (Continued)

AmI has also benefited from the convergence of three key technological components, which underlie it:

- 1) pervasive computing,
- 2) ubiquitous communication,
- 3) intelligent user interfaces (IUIs).

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1.1 Pervasive Computing

- According to the Centre for Pervasive Computing, this technology is defined as a computing environment where information and communication technology are “everywhere, for everyone, at all times.”
- The term “ubiquitous computing” was coined by Mark Weiser (1991), who envisioned “omnipresent computers” that serve people in their everyday lives, both at home and at work.

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1.2 Ubiquitous Communication

Ubiquitous communication aims at ensuring flexible and omnipresent communication between interlinked computer devices (Raisinghani et al., 2004) via:

- wireless local area networks (W-LANs),
- wireless personal area networks (W-PANs),
- wireless body area networks (W-BANs),
- Radio Frequency Identification (RFID).

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IUIs (Continued)

- With IUIs, AmI remains in the background and is virtually invisible to the user.



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1.3.1 Autonomy and Freedom Involving AmI

- We can ask whether human autonomy and freedom will be enhanced or diminished as a result of AmI technology.
- AmI's supporters suggest humans will gain more control over the environments with which they interact because technology will be more responsive to their needs.
- Brey notes a paradoxical aspect of this claim, pointing out that "greater control" is presumed to be gained through a "delegation of control to machines."

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1.3 Intelligent User Interfaces (IUIs)

- *Intelligent User Interfaces* (or *IUIs*) have been made possible by developments in AI.
- Brey (2005) notes that IUIs go beyond traditional interfaces such as a keyboard, mouse, and monitor.

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Ethical and Social Issues Affecting AmI

- Three kinds of ethical/social issues affecting AmI:
 1. freedom and autonomy;
 2. technological dependency;
 3. privacy, surveillance, and the "Panopticon."

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Autonomy and Freedom

- Brey describes three ways in which AmI may make the human environment more controllable, noting that it can:
 - i. become more responsive to the voluntary actions, intentions, and needs of users;
 - ii. supply humans with detailed and personal information about their environment;
 - iii. do what people want without having to engage in any cognitive or physical effort.

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Autonomy and Freedom

- Brey also describes three ways that Aml can diminish the amount of control that humans have over their environments, where users may lose control because a smart object can:
 - i. make incorrect inferences about the user, the user's actions, or the situation;
 - ii. require corrective actions on the part of the user;
 - iii. represent the needs and interests of parties other than the user.

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Technological Dependency

- On the one hand, IUIs could relieve us of having to worry about performing many of our routine day-to-day tasks, which can be considered tedious and boring.
- On the other hand, however, IUIs could also eliminate much of the cognitive effort that has, in the past, enabled us to be fulfilled and to flourish as humans.

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Technological Dependency (Continued)

- What would happen to us if we were to lose some of our cognitive capacities because of an increased dependency on cyberotechnology?
- Review Scenario 12-2 (in the textbook), based on E. M. Forster's insights about what could happen to a society that becomes too dependent on machines.

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1.3 Privacy, Surveillance, and the Panopticon

- Langheinrich (2001) argues that with respect to privacy and surveillance, four features differentiate Aml from other (mostly earlier) kinds of computing applications:
 - i. ubiquity,
 - ii. invisibility,
 - iii. sensing,
 - iv. memory application.

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Privacy, Surveillance, and the Panopticon (Continued)

- Langheinrich also believes that Aml poses a more significant threat to privacy than earlier computing technologies because:
 - a) sensing devices associated with IUIs may become so sophisticated that they will be able to sense (private) human emotions like fear, stress, and excitement;
 - b) this technology has the potential to create a memory or "life-log" – i.e., a complete record of someone's past.

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Surveillance and the Panopticon

- Čas (2004) notes that in Aml environments, no one can be sure that he or she is not being observed.
- Because of Aml environments, it may be prudent for a person to assume that information about his or her presence (at any location and at any time) is being recorded.

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Surveillance and the Panopticon

- Čas believes that it is realistic to assume that any activity (or inactivity) about us that is being monitored in an AMU environment may be used in any context in the future.
- So, people in Aml environments are subject to a virtual “panopticon.”
- Review Scenario 12-3 (in the textbook), based on Bentham’s idea of the *Panopticon*.
- Which Aml-related threats does it anticipate?

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Table 12-1 Ambient Intelligence

Technological Components	Ethical and Social Issues Generated
Pervasive Computing	Freedom and Autonomy
Ubiquitous Communication	Privacy and Surveillance
Intelligent User Interfaces (IUIs)	Technological Dependence

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